

Ratio: initial assessment — answers

Match each answer to its question.

1. Share £35 between A and B in the ratio 2:5. How much does A receive?

Answer: £10
7 parts = £35; 1 part = £5; A receives $2 \times £5$.

2. Red and blue counters are in the ratio 3:4. There are 12 red counters. How many blue counters are there?

Answer: 16 blue counters
3 parts = 12; 1 part = 4; blue = 4×4 .

3. Two ribbons have lengths in the ratio 2:7. The longer is 25 cm longer than the shorter. Find the shorter length.

Answer: 10 cm
5 extra parts = 25 cm; 1 part = 5 cm; shorter = 2×5 cm.

4. Adults and children are in the ratio 5:8. There are 39 people altogether. How many more children than adults are there?

Answer: 9 more children
13 parts = 39; 1 part = 3; difference = $(8 - 5) \times 3$.

Ratio: answers and worked method

Answers to step-by-step practice 1

	Thread 1: Guided	Thread 2: Core	Thread 3: Depth
The Full Question	An 84 ml paint mix contains red and white paint in the ratio 3:4. How much red paint is in the mix?	Notebooks and folders are in the ratio 4:7. There are 35 folders. How many notebooks are there?	Adults and children are in the ratio 2:5. There are 24 more children than adults. How many people are there altogether?
Keywords & Calculation	Total known: $3 + 4 = 7$ parts. Find 1 part, then red paint's 3 parts.	35 folders is 7 parts. Find 1 part, then 4 parts for notebooks.	24 more children is $5 - 2 = 3$ parts. Find 1 part, then 7 parts altogether.
Written Method	84 ml $\div 7 = 12$ ml per part. $12 \text{ ml} \times 3 = 36$ ml. Red White 84 ml altogether 1 part = 12 ml	Notebooks Folders Folders: 35 1 part = 5 items	Adults Children 3 extra parts = 24 people 1 part = 8 people
Ballpark Check & Math	36 ml is less than half of 84 ml. The other share is 48 ml; together they make 84 ml.	4 parts is just over half of 7 parts, so expect just over half of 35. 20 fits.	The total is 7 parts, the difference 3. Expect between 2×24 and 3×24 : 48 to 72. 56 fits.
Final Answer	Red paint: 36 ml.	20 notebooks.	56 people altogether.

Ratio: check your method

Step-by-step practice 1 • Compare your working; correct the first wrong step.

Thread 1: Guided	Thread 2: Core	Thread 3: Depth
Method 7 parts; $84\text{ ml} \div 7 = 12\text{ ml per part}$; $12\text{ ml} \times 3 = 36\text{ ml of red paint}$.	Method 35 folders represent 7 parts. $35 \div 7 = 5\text{ per part}$; $5 \times 4 = 20\text{ notebooks}$.	Method The difference is $5 - 2 = 3\text{ parts}$. $24 \div 3 = 8\text{ per part}$; $8 \times 7 = 56\text{ people}$.
If you got... 48 ml: check whether you used $12\text{ ml} \times 4$. That is white paint; red has 3 parts.	If you got... 32 notebooks: check whether you used $35 - (7 - 4)$. The difference of 3 is in parts, not individual items.	If you got... 168 people: check whether you used 24×7 . The 24 represents 3 parts, not 1; divide by 3 first.
Check Red: 36 ml; White: 48 ml. $36\text{ ml} + 48\text{ ml} = 84\text{ ml}$, and 36:48 simplifies to 3:4.	Check 20:35 simplifies to 4:7. There should be fewer notebooks than folders.	Check Adults: 16; children: 40. $40 - 16 = 24$; $16 + 40 = 56$.

Ratio: answers and worked method

Answers to step-by-step practice 2

	Thread 1: Guided	Thread 2: Core	Thread 3: Depth
The Full Question	A 96 cm ribbon is cut into pieces A and B in the ratio 5:7. Find the length of B.	Juice and water are mixed in the ratio 2:7. There are 18 litres of juice. How many litres of water are needed?	Two lengths of ribbon, A and B, are in the ratio 5:8. B is 21 cm longer than A. What is their total length?
Keywords & Calculation	Total known: $5 + 7 = 12$ parts. Find 1 part, then B's 7 parts.	18 litres of juice is 2 parts. Find 1 part, then 7 parts of water.	21 cm is $8 - 5 = 3$ parts. Find 1 part, then 13 parts altogether.
Written Method	$96 \text{ cm} \div 12 = 8 \text{ cm per part. } 8 \text{ cm} \times 7 = 56 \text{ cm.}$ A B 96 cm altogether 1 part = 8 cm	$18 \div 2 = 9 \text{ litres per part. } 9 \times 7 = 63 \text{ litres.}$ Juice Water Juice: 18 litres 1 part = 9 litres	$21 \div 3 = 7 \text{ cm per part. } 7 \times 13 = 91 \text{ cm.}$ A B 3 extra parts = 21 cm 1 part = 7 cm
Ballpark Check & Math	B is more than half of 96 cm. 56 cm fits; $40 \text{ cm} + 56 \text{ cm} = 96 \text{ cm.}$	Water is between 3 and 4 times the juice: between 54 and 72 litres. 63 fits.	The total is just over 4 times the 21 cm difference. $4 \times 21 = 84$; 91 is reasonable.
Final Answer	B is 56 cm.	63 litres of water.	91 cm altogether.

Ratio: check your method

Step-by-step practice 2 • Compare your working; correct the first wrong step.

Thread 1: Guided	Thread 2: Core	Thread 3: Depth
Method 12 parts; $96\text{ cm} \div 12 = 8\text{ cm}$ per part; $8\text{ cm} \times 7 = 56\text{ cm}$ for B.	Method 18 litres represent 2 parts. $18 \div 2 = 9$ per part; $9 \times 7 = 63$ litres of water.	Method The difference is $8 - 5 = 3$ parts. $21 \div 3 = 7\text{ cm}$ per part; $7 \times 13 = 91\text{ cm}$.
If you got... 40 cm: check whether you used $8\text{ cm} \times 5$. That is piece A; the question asks for B.	If you got... 23 litres: check whether you used $18 + (7 - 2)$. Five extra parts do not mean five extra litres.	If you got... 273 cm: check whether you used 21×13 . The 21 cm represents 3 parts, not 1.
Check $40\text{ cm} + 56\text{ cm} = 96\text{ cm}$. B is more than half of 96 cm.	Check 18:63 simplifies to 2:7. Water is between 3 and 4 times the juice: 54 to 72 litres.	Check Lengths: 35 cm and 56 cm. Difference: 21 cm. Total: 91 cm.

Ratio: answers

Independent practice 1 • Match the thread and question number.

Thread 1: Guided	Thread 2: Core	Thread 3: Depth
1. £36	1. 300 g	1. 24 m
2. 120 ml	2. 24 tickets	2. Red: 6 counters Blue: 30 counters
3. Shorter: 40 cm Longer: 56 cm	3. 20 m	3. £40
4. 30 beads	4. £63	4. 675 g
5. 250 g	5. 18 L	5. Water: 1200 ml Juice: 200 ml
6. 18 min	6. 40 min	6. 99 min

Check what is requested: one amount, both amounts, a total or a difference.

The next slide checks a method and a possible error in each thread.

Ratio: check your method

Independent practice 1 • Compare your working; correct the first wrong step.

Thread 1: Guided	Thread 2: Core	Thread 3: Depth
Method Q4: $5 - 3 = 2$ parts. $12 \div 2 = 6$ beads per part. Blue: $6 \times 5 = 30$ beads.	Method Q3: $7 - 4 = 3$ parts. $15 \div 3 = 5$ m per part. Rope A: $5 \times 4 = 20$ m.	Method Q3: $7 + 3 = 10$ parts. $\text{£}100 \div 10 = \text{£}10$ per part. Shares $\text{£}70$ and $\text{£}30$; difference $\text{£}40$.
If you got... Q4 — 60 beads: check whether you used 12×5 . The 12 extra beads are 2 parts, not 1.	If you got... Q3 — 60 m: check whether you used 15×4 . The 15 m difference represents 3 parts, not 1.	If you got... Q3 — $\text{£}70$: check whether you stopped at $\text{£}10 \times 7$. That is Ava's share; subtract Leo's $\text{£}30$.
Check Q4: Red 18, blue 30. $30 - 18 = 12$; $18:30 = 3:5$.	Check Q3: A is 20 m; B is 35 m. Difference 15 m; ratio 4:7.	Check Q3: $\text{£}70 + \text{£}30 = \text{£}100$; $\text{£}70 - \text{£}30 = \text{£}40$.

Ratio: answers

Independent practice 2 • Match the thread and question number.

Thread 1: Guided	Thread 2: Core	Thread 3: Depth
1. 70 g	1. 27 km	1. 440 g
2. 54 m	2. 210 ml	2. 25 L
3. 30 min	3. £35	3. Reading: 24 min Writing: 54 min
4. £40	4. 12 min	4. 63 cm
5. 35 flowers	5. 140 g	5. £26
6. 240 ml	6. 80 animals	6. 153 tickets

Check what is requested: one amount, both amounts, a total or a difference.

The next slide checks a method and a possible error in each thread.

Ratio: check your method

Independent practice 2 • Compare your working; correct the first wrong step.

Thread 1: Guided	Thread 2: Core	Thread 3: Depth
Method Q5: $5 - 4 = 1$ part. The 7 extra flowers are 1 part. White: $7 \times 5 = 35$ flowers.	Method Q2: $3 + 4 = 7$ parts. $490 \div 7 = 70$ ml per part. Red paint: $70 \times 3 = 210$ ml.	Method Q2: Tank A's 15 litres is 3 parts. $15 \div 3 = 5$ per part. Tank B: $5 \times 8 = 40$ litres. Difference: $40 - 15 = 25$ litres.
If you got... Q5 — 28 flowers: check whether you used 7×4 . That counts red flowers; white has 5 parts.	If you got... Q2 — 70 ml: check whether you stopped after $490 \div 7$. That is one part; red paint needs 3 parts.	If you got... Q2 — 40 litres: check whether you stopped at 5×8 . The question asks how much more, so subtract A's 15 litres.
Check Q5: Red 28, white 35. $35 - 28 = 7$; $28:35 = 4:5$.	Check Q2: Red 210 ml + white 280 ml = 490 ml; ratio 3:4.	Check Q2: $15:40 = 3:8$, and $15 + 25 = 40$ litres.

Ratio: answers

Independent practice 3 • Match the thread and question number.

Thread 1: Guided	Thread 2: Core	Thread 3: Depth
1. 24 members	1. 72 min	1. 8 min
2. 400 ml	2. 56 mm	2. £168
3. £14	3. 210 g	3. 77 animals
4. 90 min	4. 24 beads	4. 150 ml
5. 100 g	5. £9	5. 91 m
6. 72 cm	6. 825 ml	6. Oats: 500 g Raisins: 450 g

Check what is requested: one amount, both amounts, a total or a difference.

The next slide checks a method and a possible error in each thread.

Ratio: check your method

Independent practice 3 • Compare your working; correct the first wrong step.

Thread 1: Guided	Thread 2: Core	Thread 3: Depth
Method Q3: $6 - 1 = 5$ parts. $£70 \div 5 = £14$ per part. Dan gets 1 part: £14.	Method Q3: Butter's 90 g is 3 parts. $90 \div 3 = 30$ g per part. Total: $30 \times (4 + 3) = 210$ g.	Method Q2: May's £60 is 5 parts. $£60 \div 5 = £12$ per part. Total: $£12 \times (9 + 5) = £168$.
If you got... Q3 — £70: check whether you treated the £70 difference as 1 part. It represents 5 parts, so divide by 5.	If you got... Q3 — 120 g: check whether you stopped at 30×4 . That is flour only; include 90 g of butter.	If you got... Q2 — £108: check whether you stopped at $£12 \times 9$. That is Kai's share; add May's £60.
Check Q3: Dan £14, Pat £84. $£84 - £14 = £70$; $14:84 = 1:6$.	Check Q3: Flour 120 g + butter 90 g = 210 g; ratio 4:3.	Check Q2: $£108 + £60 = £168$; $108:60 = 9:5$.

Ratio: answers

Independent practice 4 • Match the thread and question number.

Thread 1: Guided	Thread 2: Core	Thread 3: Depth
1. 20 min	1. £192	1. 9 L
2. First: 5 L Second: 15 L	2. 54 plants	2. 540 g
3. 36 cm	3. 5 min	3. 45 cm
4. £84	4. 12 kg	4. 117 tickets
5. 48 chocolates	5. 6 L	5. £88
6. 240 g	6. 21 m	6. 50 min

Check what is requested: one amount, both amounts, a total or a difference.

The next slide checks a method and a possible error in each thread.

Ratio: check your method

Independent practice 4 • Compare your working; correct the first wrong step.

Thread 1: Guided	Thread 2: Core	Thread 3: Depth
Method Q3: $3 - 2 = 1$ part. The extra 12 cm is 1 part. Ribbon B: $12 \times 3 = 36$ cm.	Method Q4: $7 - 3 = 4$ parts. $16 \div 4 = 4$ kg per part. Lighter sack: $4 \times 3 = 12$ kg.	Method Q6: $7 + 2 = 9$ parts. $90 \div 9 = 10$ minutes per part. Work 70; breaks 20. Difference: 50 minutes.
If you got... Q3 — 24 cm: check whether you used 12×2 . That is ribbon A; B has 3 parts.	If you got... Q4 — 48 kg: check whether you used 16×3 . The 16 kg difference represents 4 parts, not 1.	If you got... Q6 — 70 minutes: check whether you stopped at 10×7 . Subtract 20 minutes of breaks to find how much longer.
Check Q3: A is 24 cm; B is 36 cm. Difference 12 cm; ratio 2:3.	Check Q4: Heavier sack 28 kg; lighter 12 kg. Difference 16 kg; ratio 7:3.	Check Q6: $70 + 20 = 90$ minutes; $70 - 20 = 50$ minutes.